

L2.1 Maximum and Minimum Values

§3.1 — what you need to know

The four rules

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|--|---|
| ① Ends can't be local max or min. | No neighbourhood around an endpoint — only half of one. |
| ② A single point <i>can</i> be a local max. | On a flat stretch every interior point is a local max <i>and</i> a local min. |
| ③ A peak can be <i>both</i> local and absolute. | Not exclusive. Give both labels when both are true. |
| ④ Hole or VA at the extreme value $\Rightarrow$ no extremum there. | A value the function never attains cannot be its maximum. |

Absolute compares $f(c)$ against *every* x in the domain; **local** compares it against every x *near* c . That one word is the whole difference. **Extremum** means a maximum *or* a minimum, either one — plural **extrema**.

EXTREME VALUE THEOREM

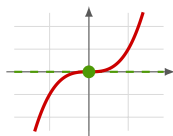
$$f \text{ cont on } [a, b] \implies \exists \text{ ABS min/Max values for some numbers on } [a, b]$$

Two hypotheses, and **closed** carries as much weight as *continuous*. The conclusion is $\exists$ and nothing more — the two values are **attained**. It does not say what they are, where they are, or that there is only one of each. Break either hypothesis and an extremum can vanish. **Bounded is not the same as attained**, and that gap is the entire content of the theorem.

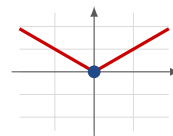
FERMAT'S THEOREM

$$\text{loc min/Max at } c \text{ and } f'(c) \exists \implies f'(c) = 0$$

One direction only, and both hypotheses are load-bearing. The two ways it gets misread are exactly these two pictures.



- ① **The converse fails.** $f(x) = x^3$ has $f'(0) = 0$ and *no* extremum — the curve crosses its horizontal tangent.



- ② f' **can fail to exist at one.** $f(x) = |x|$ is local *and* absolute min at 0, and $f'(x) = 0$ would never find it.

CRITICAL NUMBERS

c in the domain of f , and $f'(c) = 0$ or $f'(c)$ DNE

The critical numbers are the possible locations of the local min/Max. Fermat guarantees every local extremum sits at one, so they are the complete list of places worth checking.

Possible, not actual — a shortlist, not an answer. f' tells you where to look, never what you found.

When f' comes out as one fraction, read both halves off it: a fraction is zero where its numerator is, and fails to exist where its denominator is.

THE CLOSED INTERVAL METHOD

For a **continuous** f on a **closed** $[a, b]$:

- ① Find the critical numbers of f in (a, b) and evaluate f at each one.
- ② Also evaluate the function at the endpoints, $f(a)$ and $f(b)$.
- ③ The largest of those values is the absolute maximum; the smallest is the absolute minimum.

Why the list is complete. EVT guarantees the two values exist; Fermat guarantees any *local* extremum is at a critical number; anything else has to be at an endpoint. There is nowhere else for them to be — so **no test is needed at the end**, and the largest number on the list simply *is* the maximum.

WATCH OUT

~~✗ the endpoint is a local max~~ → rule ① — ends can be absolute, never local

~~✗ $f'(c) = 0$, so c is a minimum~~ → watch-out ① — x^3 at 0 is a horizontal tangent and nothing else

~~✗ no derivative there, so no extremum~~ → watch-out ② — $|x|$ at 0 is local *and* absolute

~~✗ the maximum is 3, at the open circle~~ → rule ④ — 3 is never attained, so it is a bound, not a maximum

~~✗ EVT tells me the maximum value~~ → it gives *existence* only — the Closed Interval Method gives the number

~~✗ every critical number is an extremum~~ → it is a candidate; the list is where the search starts